J-space, Part 2: The Model Matrix

From lab result to publishable finding — living handout

karlb-dev*

Campaign started 2026-07-27 — preregistered before data (`27c6d3c`)

1 Where Part 1 left off, and what Part 2 must decide

Part 1 (Lab 37) replicated the descriptive geometry of the Anthropic workspace paper on `Olmo-3-32B-Think` at $\sim$10$\times$ thinner capacity, found the paper's static causal dissociation *null* on both OLMo and `Qwen3.6-27B` under energy-matched instruments, established per-item frozen J-ablation as a control-clean cross-model causal handle on retrieved content ($-2.9/-2.4$ nats), and measured a foil-calibrated 46-step workspace-ahead-of-text lead. Part 2 adjudicates *why the dissociation is missing*:

	hypothesis	discriminator
H1	externalization (think-training moved the workspace into tokens)	Workstream A: matched-pretraining sibling `Olmo-3.1-32B-Instruct`, Qwen matrix completion, `gemma-4-31B-it`
H2	occupancy (live J-space $\approx$ the output stream itself)	Workstream D: output-occupancy index across the matrix
H3	training-lab specifics	unfalsifiable; shrunk by exhausting the rest
H5	residual instruments	B (frozen-logit, fit-size, Dolma corpus, dose/persistence selection), C (load-demanding tasks, paper's own sets)

Standards: $n \geq 60$ headline cells, two seeds on decisive grids, bootstrap CIs, BH-FDR across the matrix, preregistered directional predictions (`preregistration.md`).

2 Workstream A

2.1 A0 — does the J-dictionary survive post-training?

The part-1 Think lens, applied unchanged (donor J, recipient unembedding) to `Olmo-3.1-32B-Instruct` — same base model, different post-training. Preregistered gate: probes $\geq$15/21 and multihop pass@1 $\geq 0.85\times$ donor = 0.2408.

Verdict: `TRANSFER_FAIL`, informative. The readout basis is static across post-training (unembed row cosine median 0.9984; final-norm gain cosine 1.000), direct-fact probes transfer exactly (17/21 both), the shortlist transfers — what post-training blunted is specifically the *rank-1 sharpening*, the very component that was the J-lens's distinctive advantage on Think (+41% relative pass@1 there, +8.5% here). The prereg FAIL rule fires: fit the sibling's own 120-prompt lens. Recovery to $\sim$0.28 $\Rightarrow$ mid-band J-map drift under post-training; no recovery $\Rightarrow$ Instruct resolves the silent bridge more weakly — itself an H1-flavored datum.

*Draft campaign handout; grows a subsection per banked phase. Part-1 handout: `jspace/handout/olmo32b_jspace_handout.pdf`.

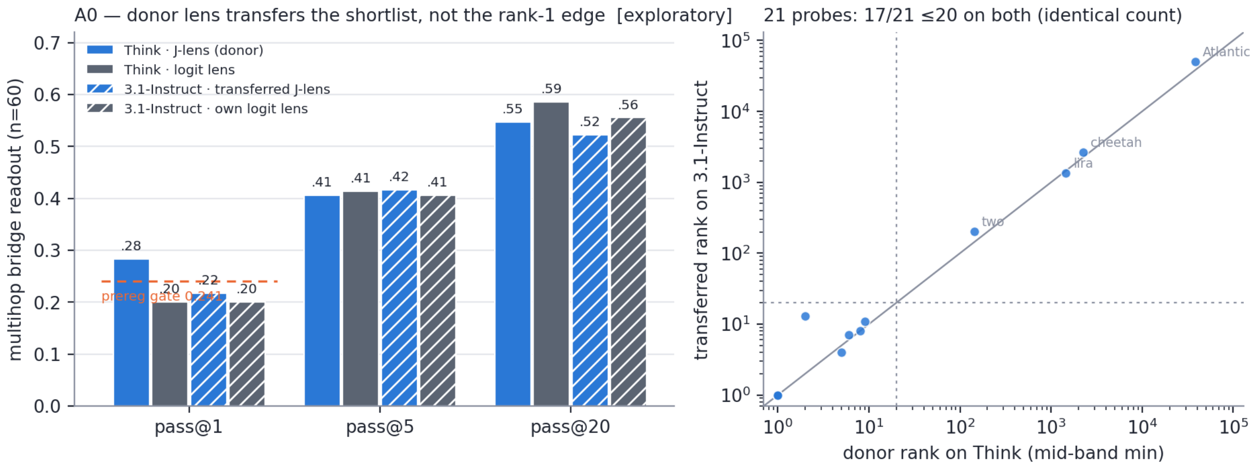

Figure 1: **A0 transfer gate.** Left: the transferred lens reads the bridge-entity *shortlist* at donor level (pass@5 0.417, pass@20 0.522 vs donor 0.41/0.55) but loses the rank-1 edge (0.217 vs 0.283, below the preregistered floor, dashed). Right: per-probe mid-band ranks — 17/21 $\leq$20 on *both* models, same items missing.

3 Workstream B

3.1 B3 — does the causal handle need the Jacobian? [exploratory pilot]

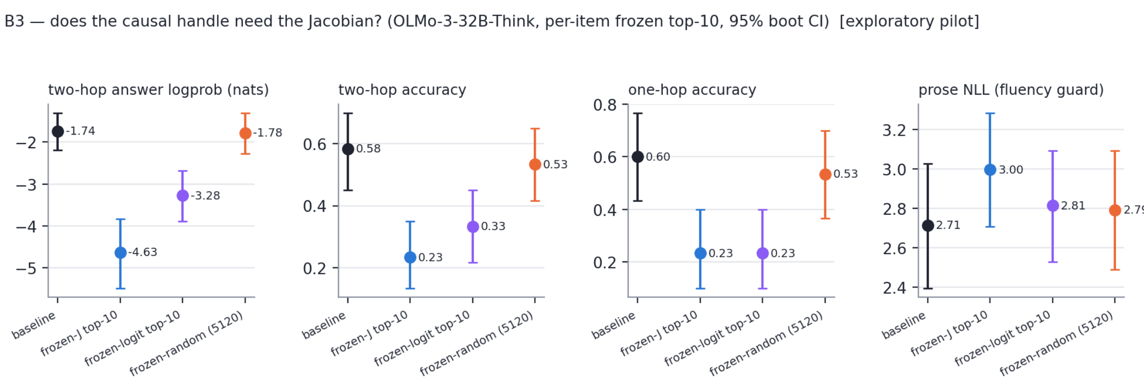

Figure 2: **B3.** The frozen-*logit* dictionary (no Jacobian, violet) reproduces ∼53% of frozen-J's two-hop logprob deletion (-1.54 vs -2.90 nats) and the *identical* one-hop collapse (0.23=0.23), while the random twin sits on baseline. Output-aligned directions alone carry much of the part-1 marquee effect; the geometry control family (R3) adjudicates the rest.

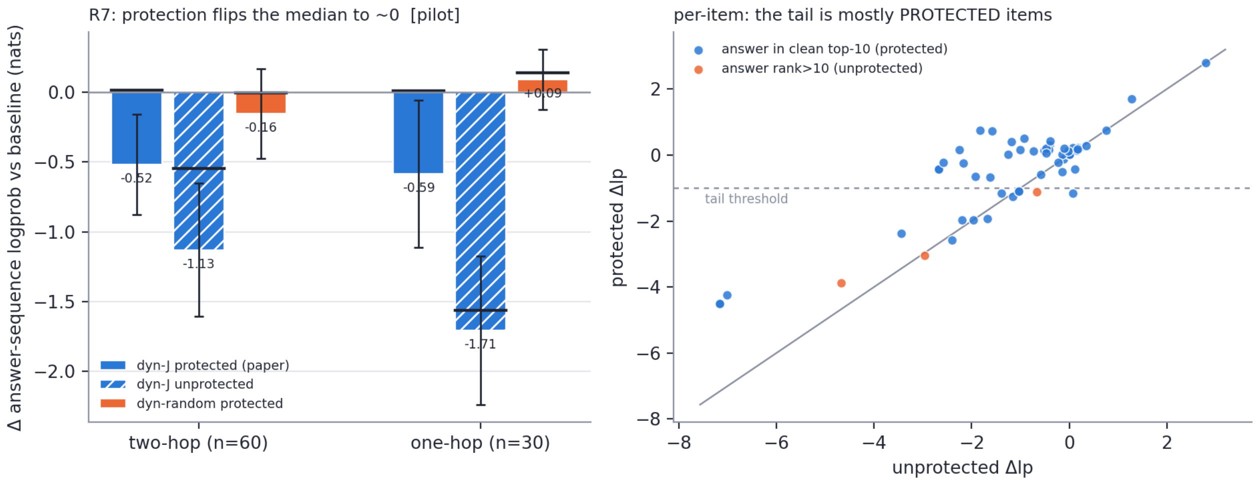

Figure 3: **R7 pilot (Think).** Left: with the paper's clean-output top-10 protection, the *median* item is untouched (two-hop +0.02, one-hop +0.01 nats) while the unprotected twin deletes hard (−0.55/−1.56 median) — part-1's live-ablation damage was mostly output-stream deletion (H2, causal form). Right: a J-specific tail survives protection (16/60 two-hop items lose > 1 nat; protected-random on the same items: +0.34), and 13 of those 16 have the answer at clean rank 1–3 — *explicitly protected yet still deleted*: indirect J-content damage (bridge-entity reading), the confirmatory grid's prime target.

4 Workstream R — assay repair (the addendum's mandate)

4.1 R7 pilot — the paper's output-protected ablation, run faithfully

4.2 R2 — capacity under the paper's own estimator

4.3 R7 cross-model + R2-Qwen — the campaign headline (pilot)

Reproducibility

Run dir `special-lab-1/part2_20260727/` (metrics per model slug, lenses, logs); code + results mirrored to `interpretability/jspace/part2/` on branch `interp_jspace_part2`; every figure regenerates from metrics via `scripts/p2fig.py`; master dataset `report/matrix_master.csv`.

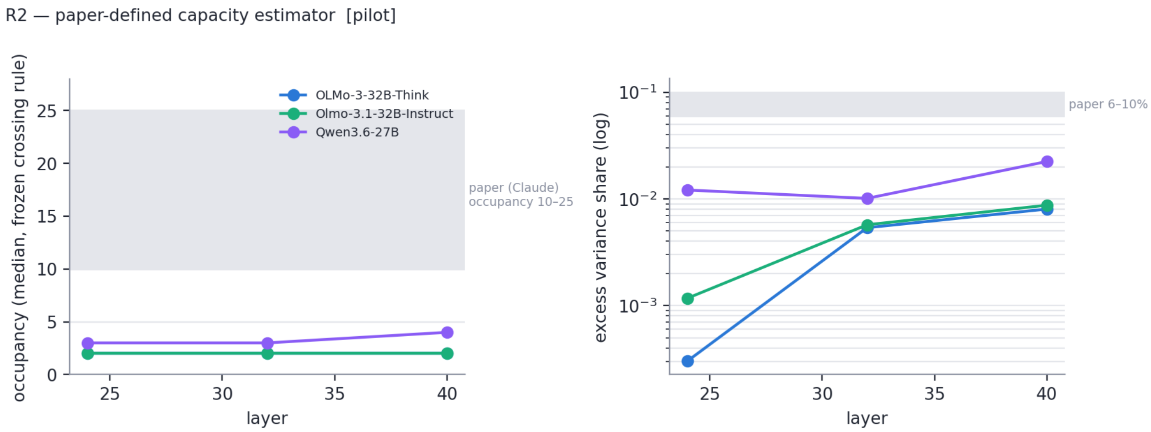

Figure 4: **R2 pilot.** Marginal-gain occupancy (frozen crossing rule, 3 random control dictionaries) and excess variance share at median occupancy. Both OLMo siblings: occupancy median 2, excess $\leq 0.9\%$ — far below the paper's Claude band (10–25, 6–10%), identical across the post-training pair. The Qwen recompute under this same estimator is the decisive cross-model capacity cell.

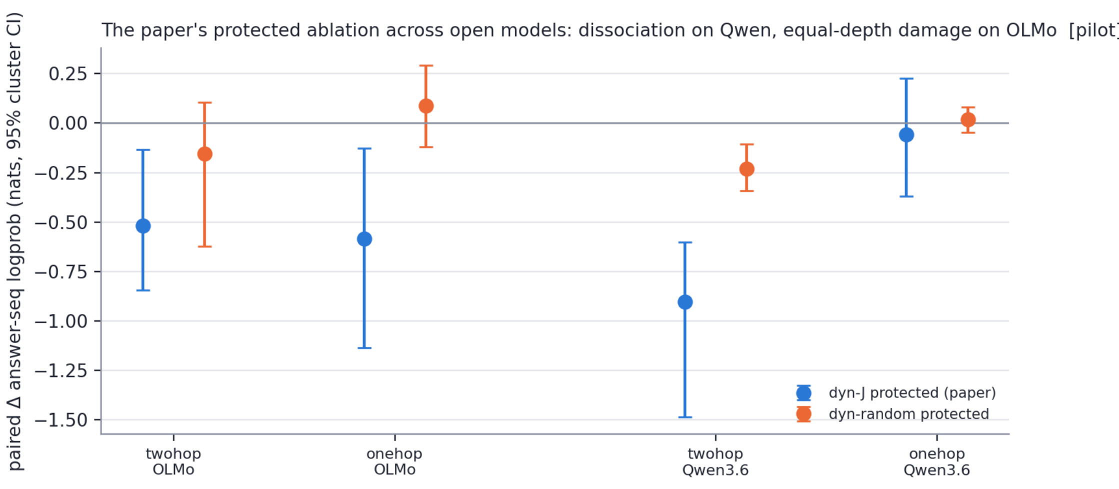

Figure 5: **Capacity and causal signature co-vary.** Under the identical protected protocol (paired family-clustered CIs): Qwen3.6-27B shows the paper's dissociation shape — two-hop -0.91 $[-1.49, -0.60]$, one-hop -0.06 [straddles 0] — while OLMo-3-32B-Think shows equal-depth content damage ($-0.52/-0.59$, both excl. 0, plus the indirect protected tail). The same models order the same way on paper-defined occupancy (3–4 vs 2, excess 1.0–2.2% vs $\leq$0.9%), and BOTH sit an order of magnitude below the paper's Claude band (10–25, 6–10%) — with lens fit size ruled out (Qwen used the published n=1000 lens). Lens robustness: two independent 120-prompt Instruct fits agree at dictionary row-cos 0.993 and frozen-selection Jaccard 0.806.